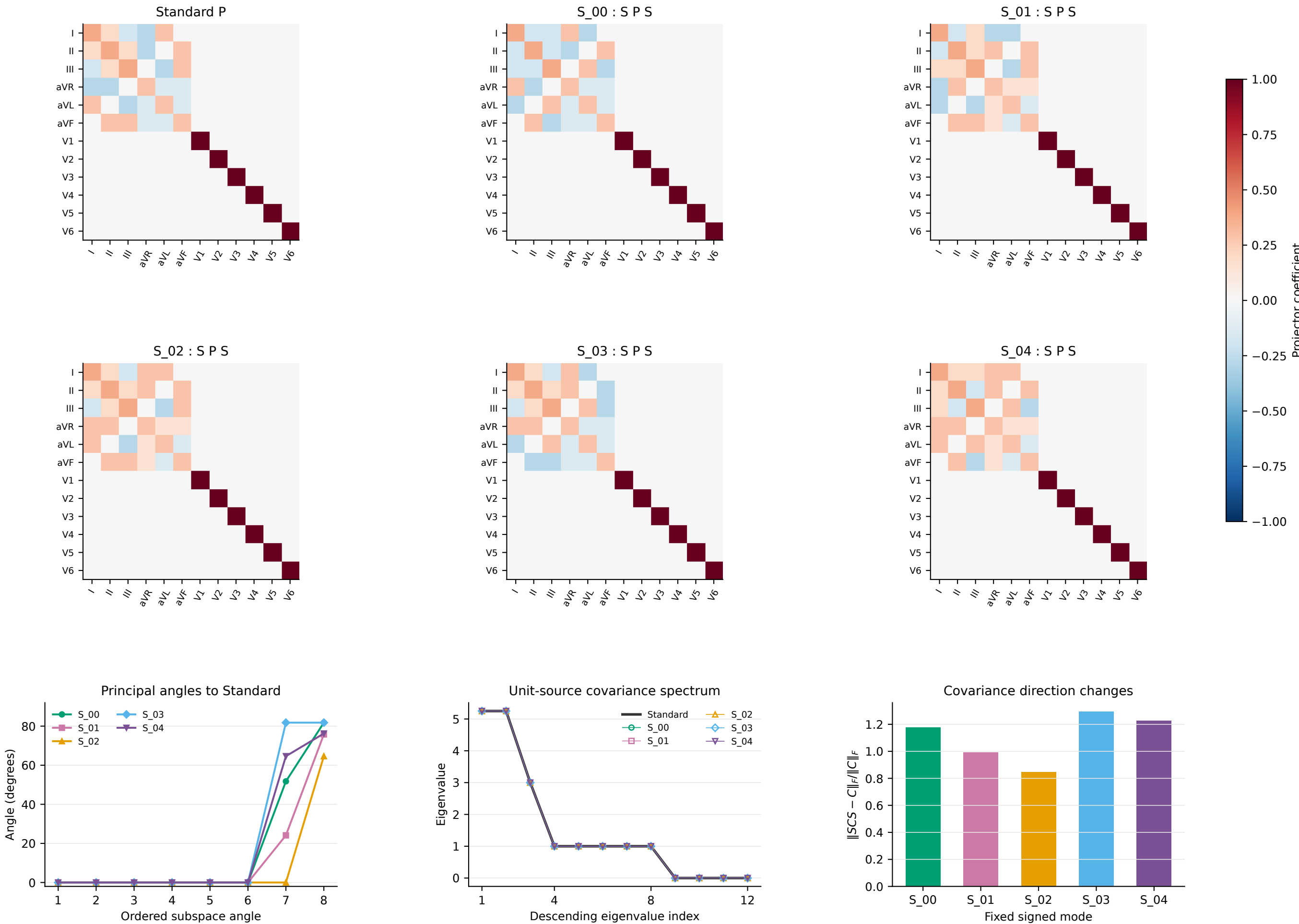


Exact subspace geometry | rotated directions, unchanged spectrum



Top: actual 12 x 12 orthogonal projectors in ECG-lead coordinates, with a shared scale. Bottom: principal angles and covariance descriptors from actual matrices. $C = A A^T$; signed covariance = $S C S$. Spectrum curves overlap by design; principal-angle roundoff can produce numerical-near-zero angles. No literal low-dimensional plane or physical hardware is implied. Standard rank 8 includes six unconstrained precordial coordinates plus ideal limb relationships.